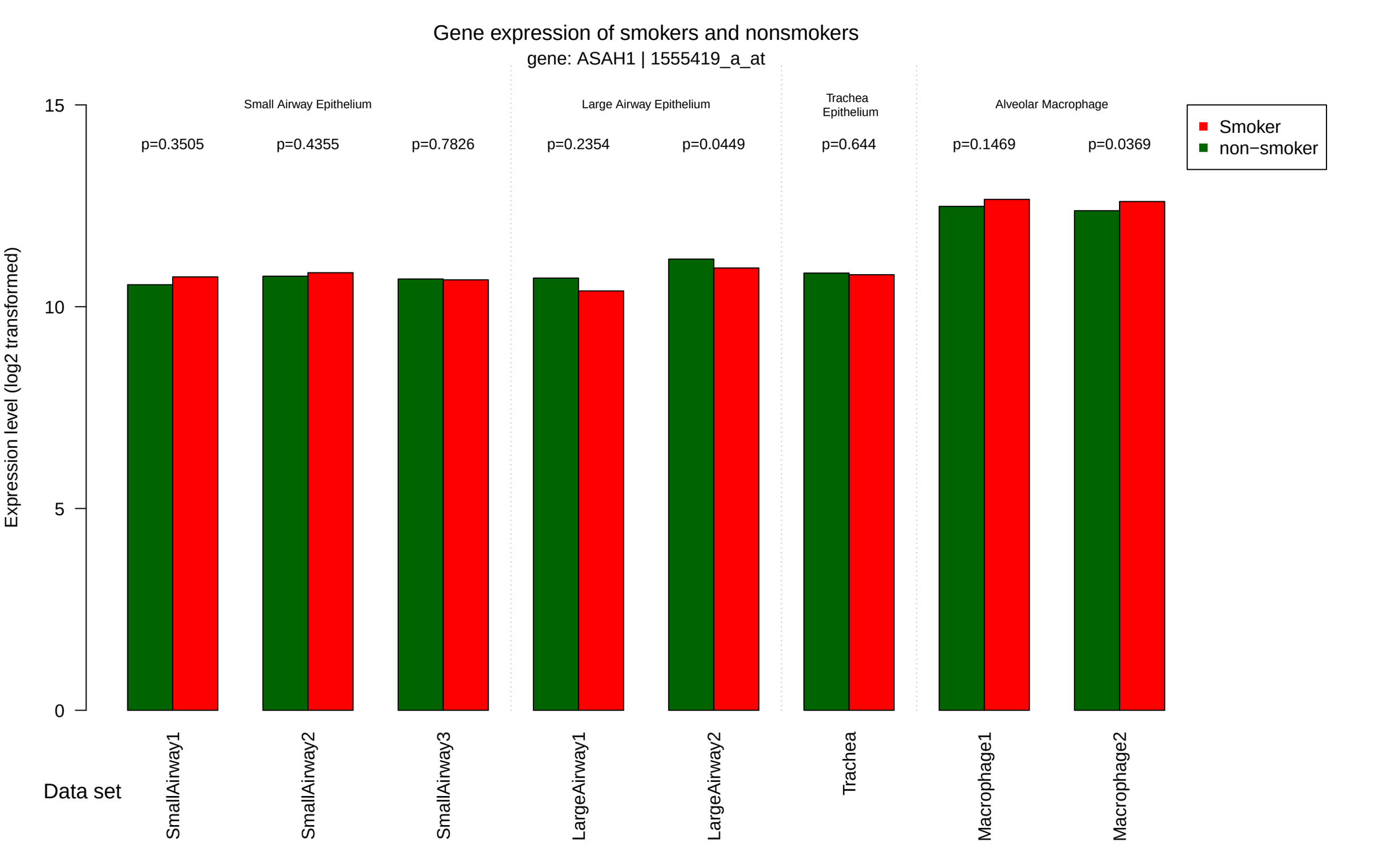
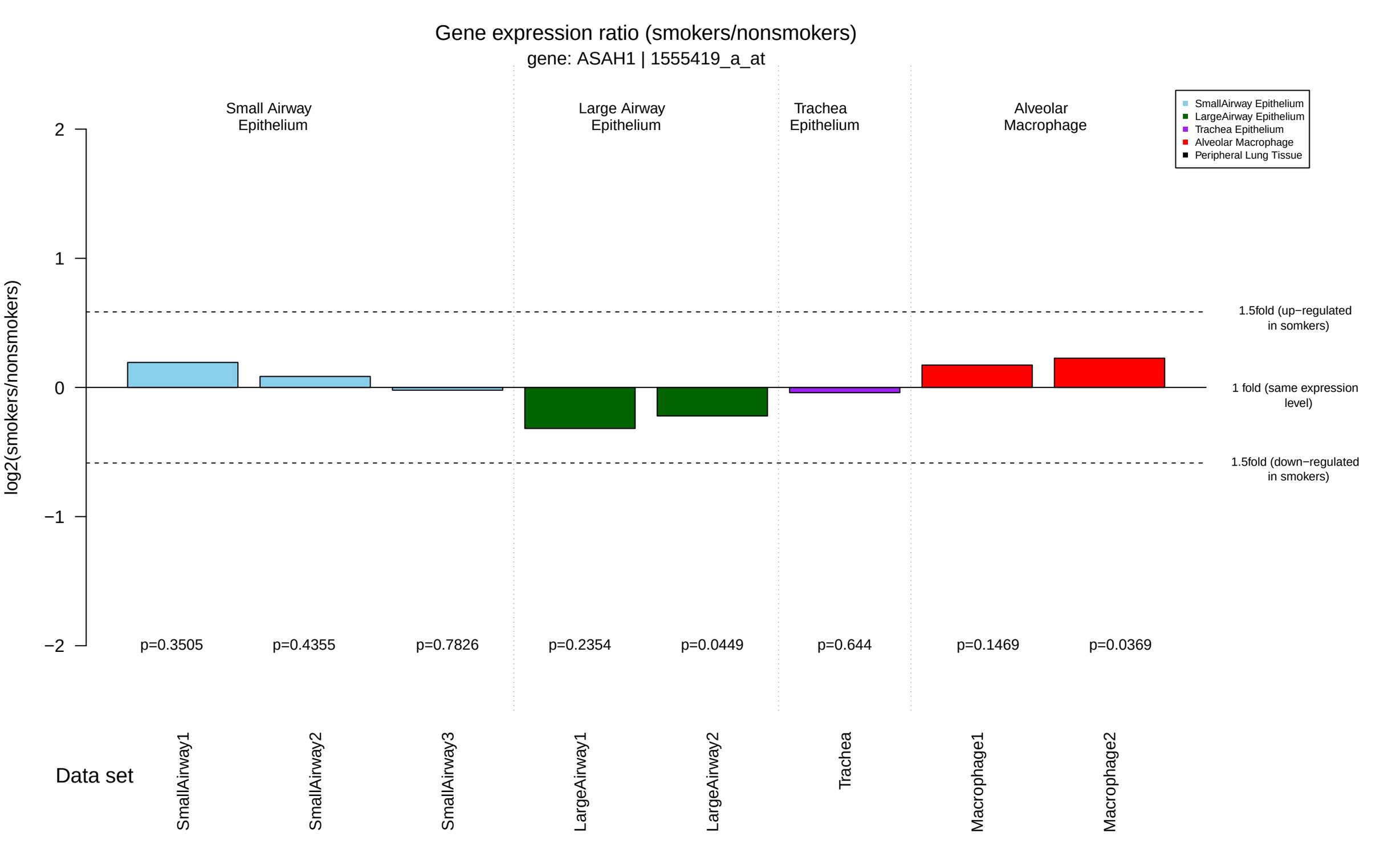
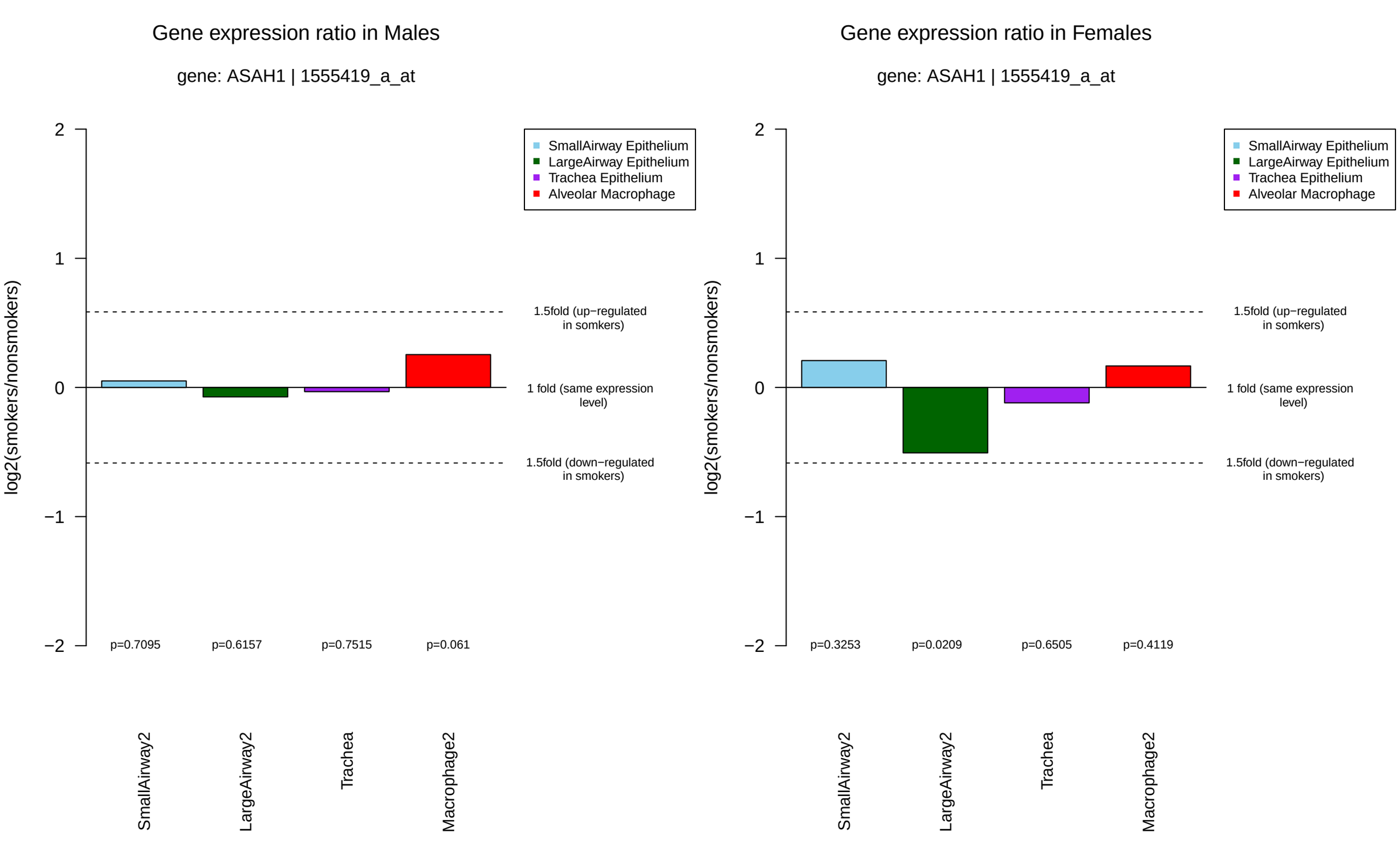
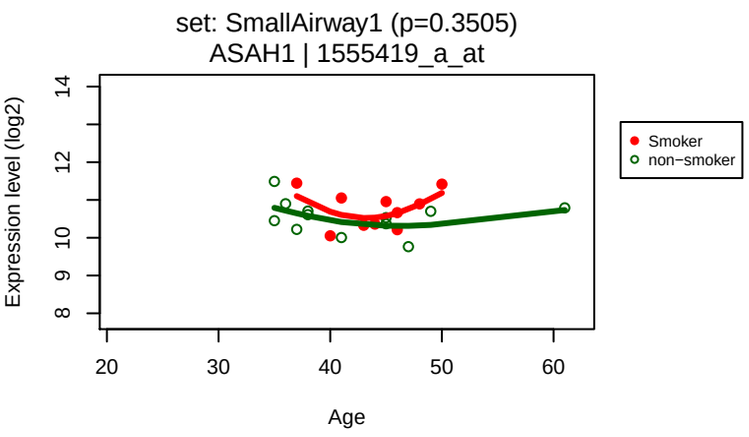
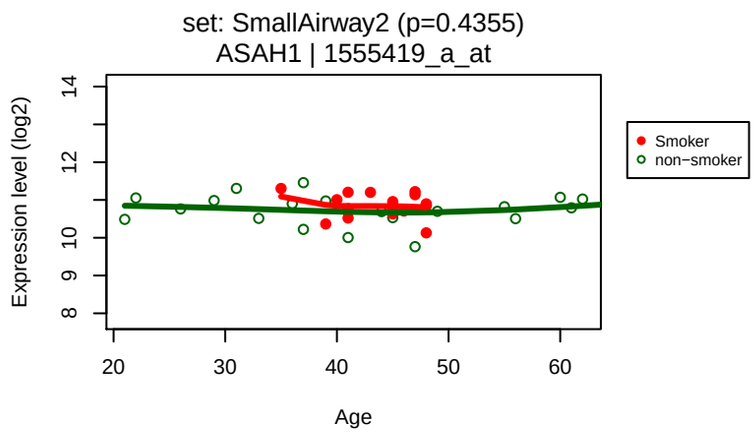
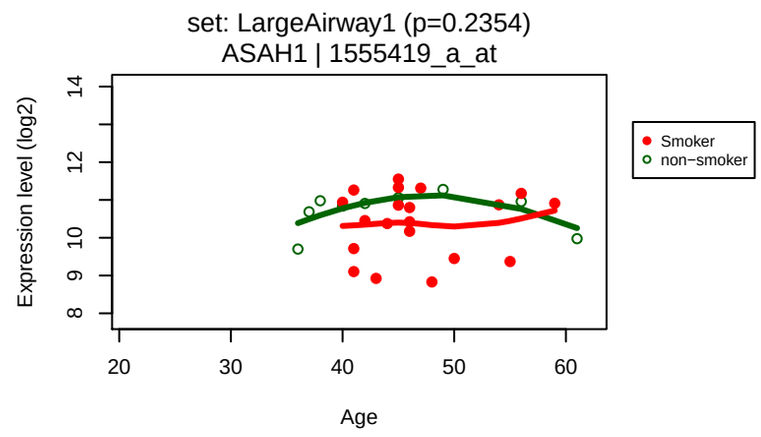

Gene expression of smokers and nonsmokers
gene: ASAH1 | 1555419_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3505
p=0.4355
p=0.7826
p=0.2354
p=0.0449
p=0.644
p=0.1469
p=0.0369
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

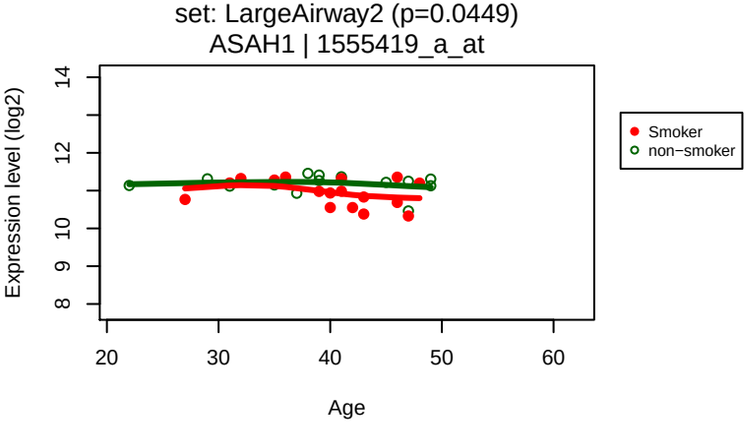

Gene expression ratio (smokers/nonsmokers)
gene: ASAH1 | 1555419_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3505
p=0.4355
p=0.7826
p=0.2354
p=0.0449
p=0.644
p=0.1469
p=0.0369
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

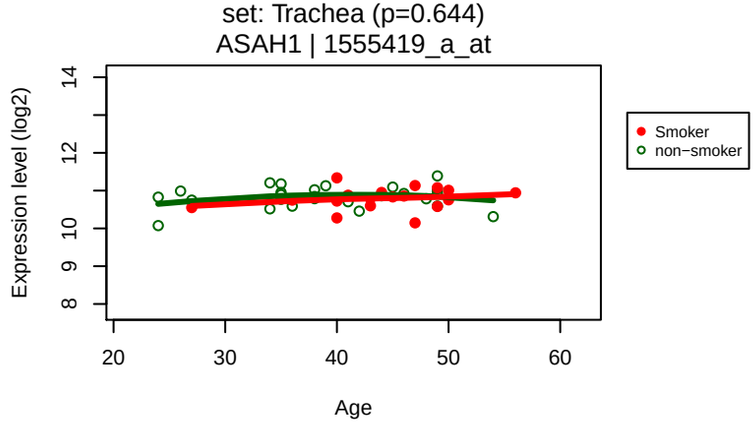

Gene expression ratio in Males
gene: ASAH1 | 1555419_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7095
p=0.6157
p=0.7515
p=0.061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASAH1 | 1555419_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3253
p=0.0209
p=0.6505
p=0.4119
SmallAirway2
LargeAirway2
Trachea
Macrophage2

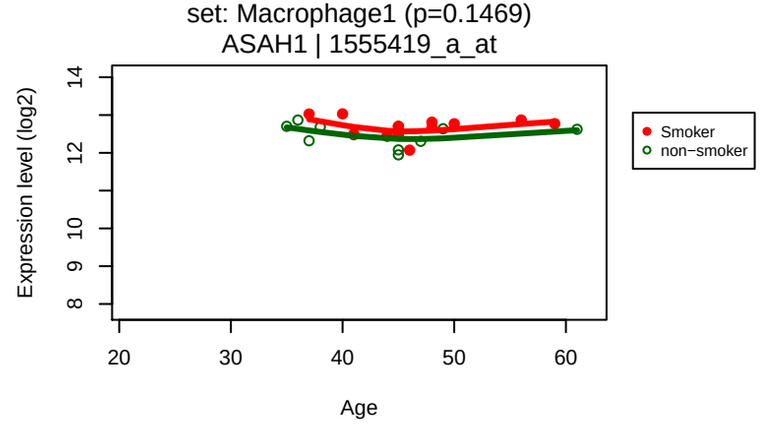

set: SmallAirway1 (p=0.3505)
ASAH1 | 1555419_a_at
Expression level (log2)
Age
Smoker
non−smoker

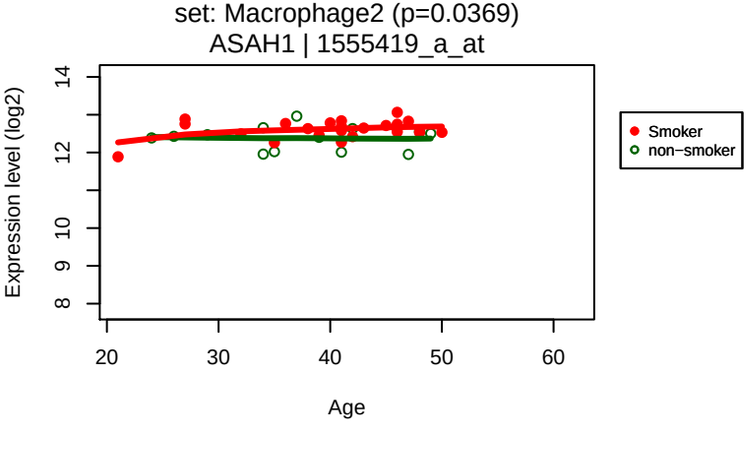

set: SmallAirway2 (p=0.4355)
ASAH1 | 1555419_a_at
Expression level (log2)
Age
Smoker
non−smoker

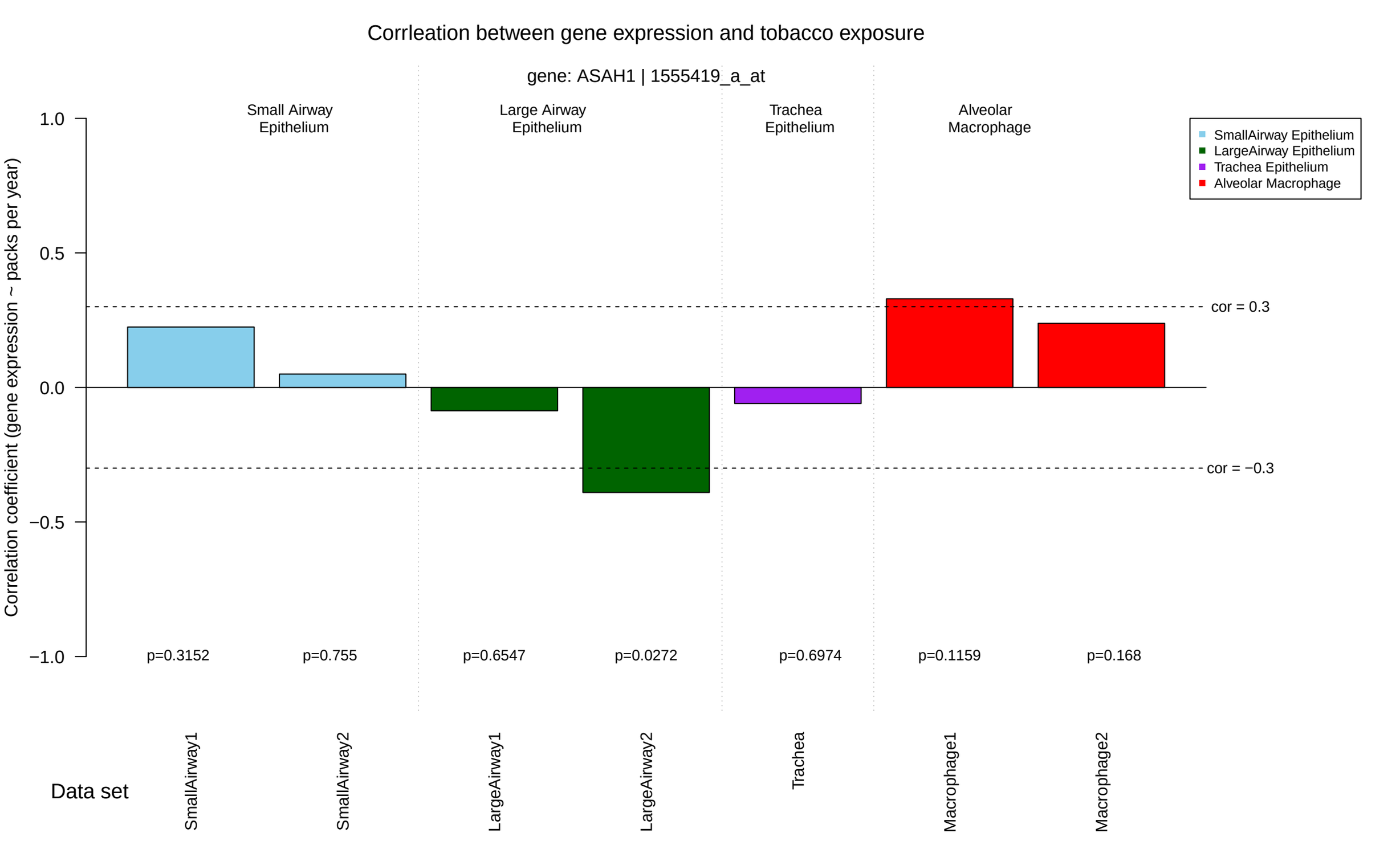

set: LargeAirway1 (p=0.2354)
ASAH1 | 1555419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0449)
ASAH1 | 1555419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.644)
ASAH1 | 1555419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1469)
ASAH1 | 1555419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0369)
ASAH1 | 1555419_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASAH1 | 1555419_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3152
p=0.755
p=0.6547
p=0.0272
p=0.6974
p=0.1159
p=0.168
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2